DESIGN SYSTEMS

TECHNICAL DATA SHEET LIVIUS-T
HIGH-PERFORMANCE SHRINKAGE-FREE SELF-COMPACTING MORTAR

DESCRIPTION

LIVIUS-T redefines industry standards as an Ultra High Performance Concrete (UHPC), a cutting-edge material that combines extraordinary strength, flexibility, hardness and unrivalled durability. Thanks to its unique formula, it eliminates the problem of expansion joints on surfaces of hundreds of square metres, providing homogeneous, flawless floors.

This mineral screed, designed for demanding professionals, is the ideal choice for creating floors with a distinctive style, either with a uniform finish or with exposed aggregates, from micro Terrazzo to terrazzo with large decorative elements. Made with minerals and aggregates of the highest quality, LIVIUS-T can be ground and polished to achieve the perfect finish, giving every project an exclusive, refined look.

Creative Freedom Without Limits

With LIVIUS-T, you customise every detail:

- Choose the colour of LIVIUS-T to harmonise it with your project.
- Opt for fine aggregates (<1 mm) for elegant, uniform surfaces, or combine stones of different sizes and colours to create unique, artistic patterns.
- Design flooring solutions that express your architectural vision without compromise.

Ease of application and optimal workability

The advanced formula of LIVIUS-T is designed to ensure extended workability (30-90 minutes), allowing applicators to achieve perfect results with the time needed to spread, level and perfect every detail.

Fields of application

The versatility of LIVIUS-T makes it ideal for a wide range of projects, from elegant residential floors to high-performance commercial and industrial floors. Thanks to its exceptional resistance to abrasion and wear, it is perfect for:

- Public and commercial spaces: schools, museums, shops, restaurants, hotel lobbies, shopping centres and airports.
- Heavy-duty industrial applications: warehouses, logistics centres, etc.

Every project created with LIVIUS-T stands out for quality, durability and refinement, proving that beauty never has to sacrifice performance.

LIVIUS-T: the ideal partner for visionary architecture and uncompromising professional applications, even directly over underfloor heating.

Adaptability and Customised Performance

The mixing speed and workability of LIVIUS-T can be optimised according to the specific requirements of the project and the site conditions. Factors such as the age of the material, the size and type of aggregates, as

well as the colours and pigments used, can influence the final result. This flexibility allows applicators to choose between a more fluid or a drier consistency, adapting the product to individual needs.

The technical characteristics of the material, including its final strength, may vary depending on the conditions of use and the mixing speed. Locally sourced materials may be used, provided they are clean, dry and free from impurities. This approach not only improves logistical efficiency, but also encourages aesthetic and functional customisation.

Environmental Benefits: Sustainable, Innovative Floors

Choosing LIVIUS-T means opting for a sustainable solution, capable of reducing CO2 emissions into the air, thanks to the lower firing temperature of the minerals used to make LIVIUS-T.

SUBSTRATE PREPARATION

Correct preparation of the base is essential to achieve excellent results with LIVIUS-T. The floor surface, whether concrete or another material, must be clean and free from dust, grease and loose particles. The substrate must guarantee a minimum tensile strength of 1,5 MPa, be dry and have a moisture content not exceeding 4%, with no risk of capillary rising damp.

In case of doubt about the moisture level, the use of specific primers is recommended to ensure optimal application. Any cracks in the base must be repaired, and weak or unsuitable surfaces must be removed or restored so that they effectively support the covering.

To ensure a uniform surface:

- Grind or shot-blast the base, then vacuum to remove all particles.
- Apply the primer appropriate to the substrate, broadcast with 0,4 to 1,8/2,0 mm sand to saturation, and allow it to dry for a maximum of 24 hours, depending on weather conditions. After drying, vacuum the surface to remove any residual sand.

MIXING: A Process Optimised for Maximum Effectiveness

To ensure uniform, professional results, the use of horizontal mixers or traditional concrete mixers is recommended when preparing LIVIUS-T. When LIVIUS-T and the aggregates are added separately, it is essential to add them simultaneously or alternately.

When mixing several bags:

- Alternate one bag of aggregates with one bag of base to obtain a perfectly balanced mix.

By following this method, the material reaches an optimal consistency, reducing the risk of errors or irregularities in the final result. The amount of water required for one bag of LIVIUS-T is approximately 14% of the product weight, depending on the absorption of the gravel used.

APPLICATION: Precision and Continuity

Prepare the work area so as to allow continuous application, avoiding cold joints that could compromise the uniformity of the result.

- Application method: The product can be applied by hand or pumped using professional machines capable of accurately dosing the required components
- Minimum thickness: Apply LIVIUS-T mortar at a minimum thickness of 10 mm. If large aggregates are present, the thickness must be increased by at least 3-5 mm over the size of the largest aggregate.
- Levelling and finishing: Use levelling bars to place the material.

Joints: If the original base has no joints, LIVIUS-T allows application without expansion joints. However, any existing joints in the underlying floor must be respected.

CURING: Essential for Durability

During and after curing, LIVIUS-T must be allowed to breathe and must not be covered with non-breathable materials.

- Temperatures and times: Low temperatures can slow down the drying process, while high temperatures can accelerate setting.
- Traffic: Avoid any type of traffic on the floor until drying is complete, to ensure uniform, lasting strength.

POLISHING: The Art of Finishing

Grinding and polishing of LIVIUS-T can begin after 24-72 hours, depending on ambient conditions.

- Exposure of aggregates: For floors with large aggregates, grind until the aggregates are correctly exposed.
- Pores and sealing: Thanks to the new formulation of LIVIUS-T, pores rarely form on the surface during polishing. Should they occur, they must be filled to obtain a flawless finish.
- Expertise required: Polishing requires advanced skills. We recommend relying on approved installers and preferring dry polishing.

SEALING: Tailored Protection

To ensure maximum performance and resistance, the floor must be sealed according to the type of use and the expected level of traffic. We recommend consulting our technical department and relying on a qualified applicator to choose the sealer best suited to the context.

DRYING

After the application and sealing of LIVIUS-T, it is essential to let the floor dry for at least 5-7 days before cleaning with water. This ensures complete stabilisation and a lasting result.

TIPS FOR A SUCCESSFUL APPLICATION

- Colour uniformity: LIVIUS-T floors may show slight colour variations. To minimise these differences, the product must be applied continuously, without prolonged interruptions, and using a uniform technique.
- Additional materials: Do not add lime, cement, gypsum or other materials to the LIVIUS-T mix.
- Rising damp: Avoid use on substrates subject to rising damp; in case of doubt, consult the supplier.
- Temperature control: During storage and installation, keep the materials between 5°C and 35°C.

DESIGN SYSTEMS

PRODUCTION, PACKAGING AND STORAGE OF LIVIUS-T

- 25 kg bags
- Storage: Store in a dry place, protected from extreme temperatures.
- Shelf life: If kept in optimal conditions, LIVIUS-T has a shelf life of 12 months from the date of production.

CRACKING

LIVIUS-T is designed to ensure high performance and exceptional strength, significantly reducing and/or eliminating the risk of cracking thanks to its expansive characteristics, because the LIVIUS-T product does not contain cement.

Values and application information (at 23°C and 50% Relative Humidity)

Flexural strength after 28 days (EN 13892-2): 10 MPa

Reaction to fire: A2-s1,d0, classified non-combustible

Compressive strength after 28 days (EN 13892-2): 60 MPa

Density of the mix kg/l: 2,2

pH of the mix: 12

Solids content %: 100

Ready for use: 7 days

Initial setting: 50 minutes

Application temperatures: +5°C / +35°C

Consumption 2,2 kg/m^2 per mm of thickness

Working time of the mix 20 minutes

Bulk density kg/l: approx. 1

Solids content %: 100

Resistance to solvents: excellent

Resistance to oils: excellent

Resistance to alkalis: excellent

Resistance to acids: moderate

Freeze-thaw, 300 cycles: the product retains more than 80% of its properties

Customs tariff code: 3824 50 90

CHARACTERISTICS of LIVIUS-T

- No shrinkage, eliminating the need for contraction joints.
- Very high flexural strength, capable of withstanding significant movements of the substrate.
- Designed to withstand stresses far exceeding those of most products, including cement-based ones.

Despite this advanced performance, it is essential that the surface is properly prepared. The condition of the substrate and the quality of the preparation directly affect the performance of the covering. Design Systems cannot be held responsible for cracks or other problems arising from poor substrates or incorrect surface preparation.

AESTHETIC VARIATIONS AND LIMITATIONS OF USE

LIVIUS-T is designed to offer a natural appearance, with visual variations that contribute to the unique charm of the floor. However, this product is not recommended for:

- Surfaces where de-icing salts are used.

TECHNICAL INFORMATION AND USER RESPONSIBILITY

Before using Design Systems products, it is essential to read carefully the up-to-date technical data sheets, bulletins and labels, available on our official website: www.cementostampato.it

WARNING:

- Do not breathe the dust: use only in well-ventilated areas and wear a NIOSH/MSHA-approved respirator (P-, N- or R-95) to minimise dust inhalation.
- Avoid contact with skin and eyes: protect eyes and skin with protective clothing, chemical-resistant gloves and waterproof boots.

SAFETY WARNINGS

- Contact with the dust or with wet mixes can cause irritation or chemical burns. In case of contact, wash immediately with plenty of soap and water.
- In case of contact with eyes, rinse immediately and thoroughly with water for at least 15 minutes and seek medical advice.

LIVIUS-T is a product intended for professional users who strictly follow the technical and safety instructions. This ensures optimal performance and protects the health of operators.

LIMITED WARRANTY

Design Systems S.r.l. declines all responsibility for consequential, incidental or special damages that may arise, directly or indirectly, from the use of the materials supplied. This exclusion ensures that our offer remains transparent, professional and focused on what we can really control: the quality of our product.

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.